

Product Requirements Document (PRD)

ACTI OS – The Digital Operating System for Amana College of Technology and Innovation

Product Name

ACTI OS

Product Vision

ACTI OS is the integrated digital platform that powers every aspect of Amana College of Technology and Innovation. It connects students, applicants, lecturers, administrators, parents, alumni, and industry partners through a single secure system that manages the entire educational journey from admission to graduation and beyond.

The goal is to make ACTI one of the most technologically advanced tertiary institutions in Nigeria by delivering a seamless, mobile-first, and AI-assisted educational experience.

Problem Statement

Most tertiary institutions operate with fragmented systems for admissions, payments, course registration, results, learning management, and communication. This creates delays, errors, poor student experience, and administrative inefficiency.

ACTI has the opportunity to build a digital-first institution from day one by creating a unified platform that eliminates paperwork, automates processes, and provides real-time visibility across academic and administrative operations.

Product Goals

- Digitize all core institutional processes.
- Provide a mobile-first experience for students and staff.
- Reduce administrative workload through automation.
- Improve transparency in admissions, finance, and academic records.
- Enable data-driven decision making.
- Support both academic and technical/vocational programmes.
- Create a scalable platform for future ACTI campuses and partner institutions.

Primary Users

Prospective Students

Apply, upload documents, track admission status, pay fees, and receive notifications.

Students

Register courses, pay fees, access learning materials, take assessments, view results, request transcripts, and communicate with lecturers and administration.

Lecturers and Instructors

Manage courses, upload materials, take attendance, create assessments, submit results, and communicate with students.

Administrators

Manage admissions, finance, academics, departments, staff, reports, and institutional operations.

Parents and Sponsors

Monitor student payments, academic progress, attendance, and official communications.

Alumni

Access transcripts, certificates, networking opportunities, and career services.

Core Modules

Admissions and Enrollment

- Online application
- Document upload
- Application tracking
- Admission decision workflow
- Acceptance fee payment
- Student ID generation

Student Information System (SIS)

- Student profiles
- Programme and department management
- Course registration
- Academic records
- Attendance
- Graduation eligibility

Finance and Payments

- Application fees
- Tuition and school fees
- Installment payment plans
- Paystack integration
- Receipts and payment history

- Financial reporting

Learning Management System

- Course materials
- Video lessons
- Assignments
- Quizzes
- CBT examinations
- Gradebook

Lecturer Portal

- Course management
- Attendance recording
- Assignment grading
- Result submission
- Student communication

Examination and Results

- CBT engine
- Continuous assessment
- Result approval workflow
- Transcript generation
- Certificate processing

Communication Hub

- Email notifications
- SMS alerts
- WhatsApp integration
- In-app announcements
- Emergency notifications

Administration Dashboard

- Enrollment analytics
- Revenue dashboard
- Attendance statistics
- Academic performance
- Department reports
- User management

Unique ACTI Features

QR Digital Student ID

Every student receives a QR-enabled digital identity used for attendance, examinations, library access, payments, and campus services.

AI Academic Assistant

Students can ask questions about courses, deadlines, regulations, fees, and campus services through an AI assistant integrated into the portal.

Skills and Employment Tracker

Tracks technical competencies, NSQ certifications, internships, and employment outcomes.

Innovation Portfolio

Each student maintains a digital portfolio of projects, prototypes, certifications, and entrepreneurial activities.

User Journey

Applicant Journey

Discover ACTI → Create account → Apply online → Upload documents → Pay application fee → Receive admission → Accept offer → Pay fees → Register courses → Become active student.

Student Journey

Log in → View dashboard → Register courses → Access learning materials → Attend classes → Take assessments → View results → Request transcript → Graduate → Join alumni network.

Non-Functional Requirements

- Mobile-first responsive design
- Accessibility compliant
- Secure authentication
- Role-based permissions
- Audit logs
- Automated backups
- High availability
- Cloud-hosted architecture
- Fast page performance

Technology Stack

- **Frontend:** Next.js + TypeScript
- **UI:** Tailwind CSS + shadcn/ui
- **Backend:** Supabase
- **Database:** PostgreSQL
- **Authentication:** Supabase Auth
- **Storage:** Supabase Storage
- **Payments:** Paystack

- **Hosting:** Vercel
- **Domain:** acti.edu.ng

Development Roadmap

Phase 1 (MVP)

- Public website
- Online admissions
- Student registration
- Payments
- Admin dashboard
- Notifications

Phase 2

- Course registration
- Lecturer portal
- Attendance
- Timetable
- Results

Phase 3

- Learning management
- CBT examinations
- AI assistant
- Parent portal
- Transcript automation

Phase 4

- Alumni network
- Internship platform
- Employer portal
- Multi-campus support
- Research and innovation management

Success Metrics

- 90% of admissions completed online
- 80% of fee payments completed digitally
- 95% reduction in paper-based student processes
- Student satisfaction score above 4.5/5
- Result processing time reduced to less than 7 days
- Administrative workload reduced by at least 50%

Long-Term Vision

ACTI OS will evolve beyond a campus portal into a **technology platform for technical and vocational education**, capable of supporting ACTI, partner institutions, satellite campuses, and government-backed TVET programmes across Nigeria.

Product Motto: *One Platform. One Campus. Unlimited Innovation.*